

MODULE 3.2: SYMMETRIC KEY ALGORITHM

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Data Encryption Standard (DES)

The Data Encryption Standard (DES) is a symmetric key block cipher used for secure data encryption. It is based on the Feistel cipher structure, designed for efficient hardware and software implementation.

Feature	Description
Type	Symmetric key block cipher
Structure	Feistel network
Developed By	IBM
Adopted By	National Institute of Standards and Technology (NIST)

It was once the most widely used encryption standard for protecting sensitive information. While DES is now considered insecure against modern brute-force attacks due to its small key size, its architectural design laid the groundwork for modern cryptography.

Key Idea of DES

DES encrypts data by:

- Splitting 64-bit plaintext into two halves
- Applying 16 rounds of Feistel operations
- Using different subkeys in each round

The same structure is used for both encryption and decryption (with reversed keys).

Feature	Description
Block Size (Both Input and Output size)	64 bits
Main Key Size	64 bits
Sub Key Size (After excluding 8 bits parity)	56 bits
Round Key	48 bits
Rounds	16 rounds

DES Encryption Flow (Short Form)

1. Input 64-bit plaintext.
2. Apply Initial Permutation (IP).
3. Split into L_0 and R_0 (32 bits each).
4. Generate sixteen 48-bit round keys.
5. Expand R from 32 bits to 48 bits.
6. XOR with round key.
7. Apply S-Box substitution.
8. Apply P-Box permutation.
9. XOR with left half.
10. Swap halves.
11. Repeat Steps 5–10 for 16 rounds.
12. Perform final swap.
13. Apply Final Permutation IP^{-1} .
14. Obtain 64-bit ciphertext.

Strengths of DES

- Fast in hardware implementation
- Strong confusion and diffusion

- Well-studied and standardized
- Uses Feistel structure (same encryption/decryption design)

Weaknesses of DES

- Small key size (56-bit), so 2^{56} possible key values → vulnerable to brute force
- Now considered insecure for modern systems
- Susceptible to exhaustive key search attacks

Importance of DES

DES Encryption Process

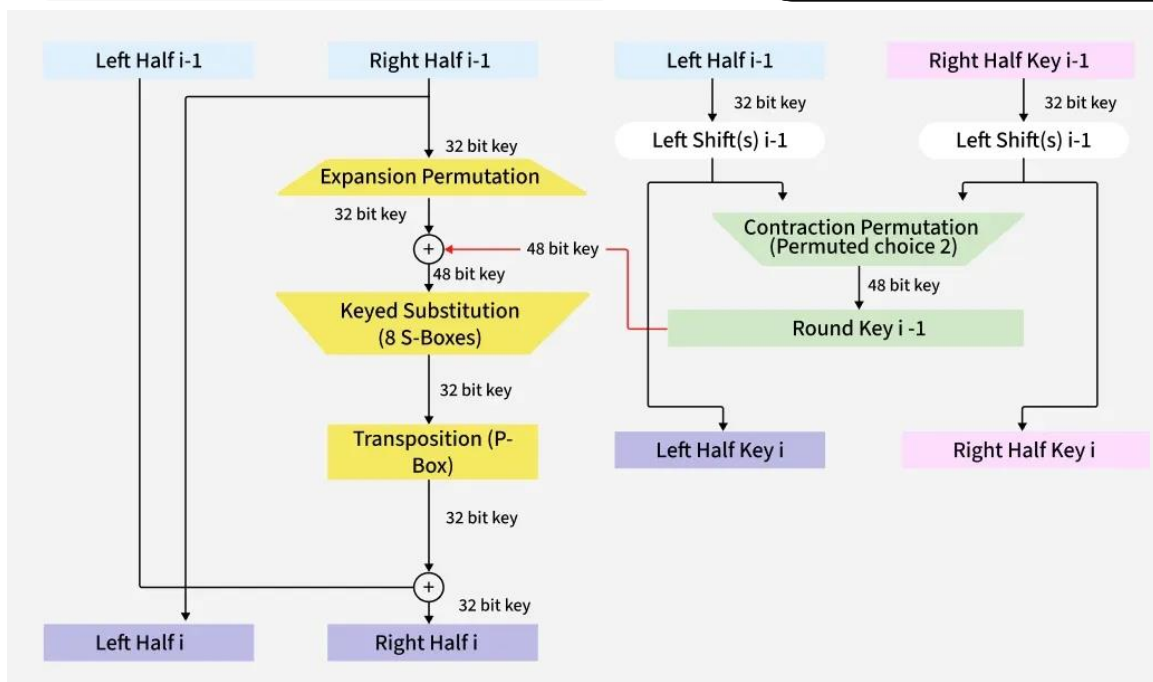
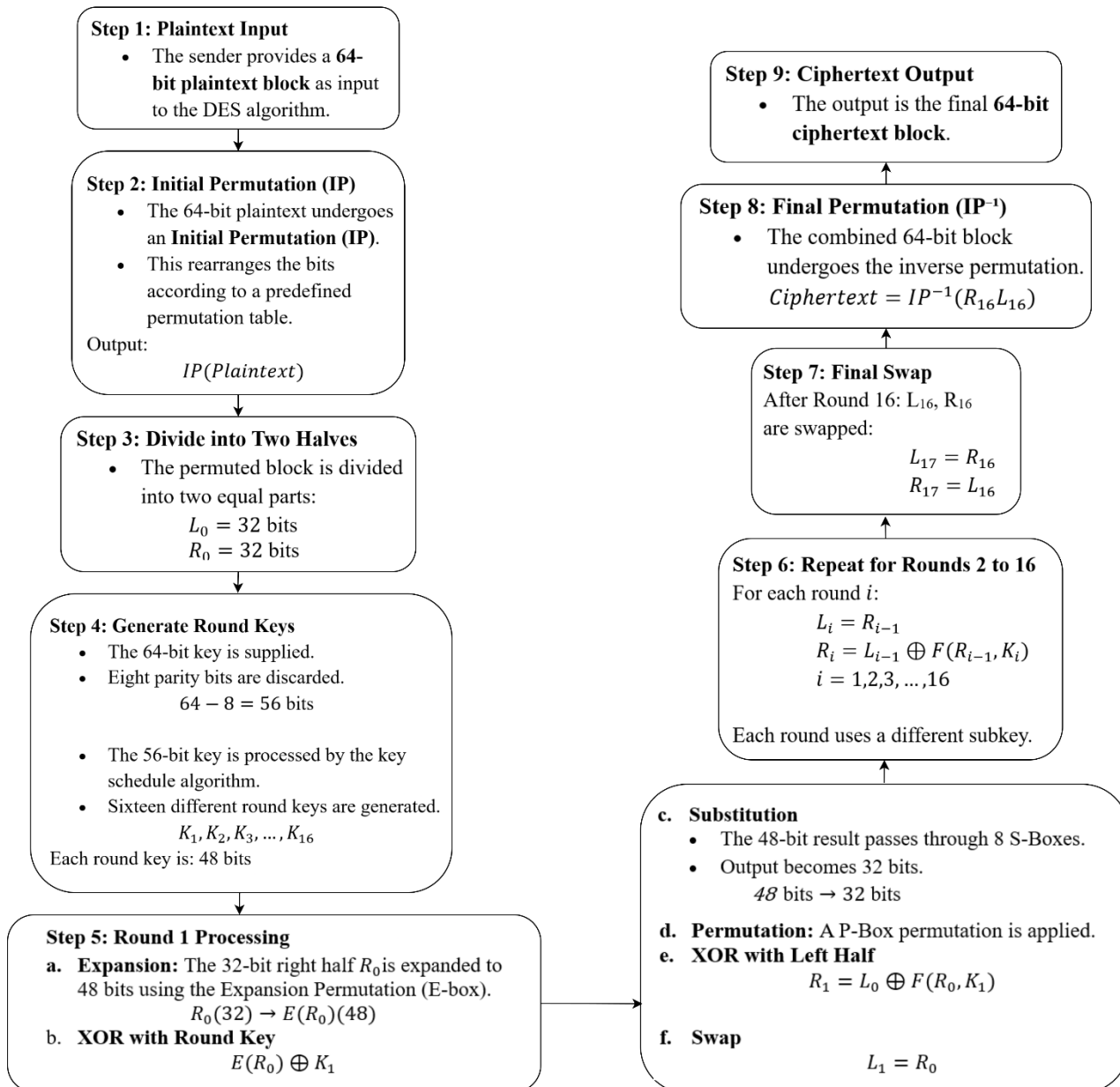


DES is historically important because:

- It was the first widely accepted encryption standard
- It introduced Feistel network in real-world cryptography
- It led to modern encryption systems like AES

Real-Life Use (Historical)

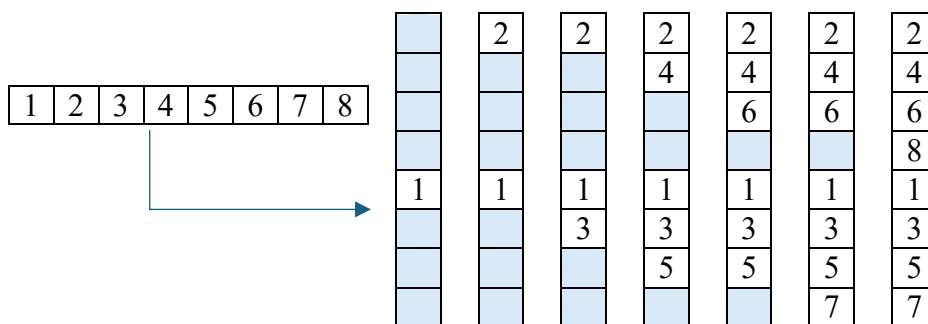
- Banking systems
- Government communication
- ATM security (earlier systems)



Step 5 in details

Initial Permutation

1	2	3	4	5	6	7	8		58	50	42	34	26	18	10	2
9	10	11	12	13	14	15	16		60	52	44	36	28	20	12	4
17	18	19	20	21	22	23	24		62	54	46	38	30	22	14	6
25	26	27	28	29	30	31	32		64	56	48	40	32	24	16	8
33	34	35	36	37	38	39	40		57	49	41	33	25	17	9	1
41	42	43	44	45	46	47	48		59	51	43	35	27	19	11	3
49	50	51	52	53	54	55	56		61	53	45	37	29	21	13	5
57	58	59	60	61	62	63	64		63	55	47	39	31	23	15	7



Inverse Initial Permutation means inversing each mapping

IP says: $58 \rightarrow 1$

then inverse says: $1 \rightarrow 58$

Decryption Process

- Same structure as encryption
- Only difference is subkeys are used in reverse order: $K_1 \rightarrow K_{16}, K_2 \rightarrow K_{15}, \dots, K_{16} \rightarrow K_1$

Key Scheduling

Key Scheduling

1. Permuted Choice 1 (PC-1):

- The original DES key is **64 bits** long.
- Every 8th bit is a **parity bit** and is discarded.
- Parity bit positions are:

$8, 16, 24, 32, 40, 48, 56, 64$

- After removing these 8 parity bits, the key size becomes:

$64 - 8 = 56 \text{ bits}$

- The 56-bit key is then permuted and divided into two halves:

$C_0 = 28 \text{ bits}, \quad D_0 = 28 \text{ bits}$

2. Left Circular Shift:

- The two 28-bit halves (C_i and D_i) are shifted left circularly before each round.
- A circular shift means that the leftmost bit removed from the front is inserted back at the right end.
- The number of shifts depends on the round number.
- Rounds 1, 2, 9, and 16 use a 1-bit shift.
- All other rounds use a 2-bit shift.
- This process generates a different key arrangement for each round, improving security.

3. Permuted Choice 2 (PC-2):

- After the left circular shifts, the two halves are combined again to form a 56-bit key.
- PC-2 selects and permutes 48 bits from these 56 bits.
- The resulting 48-bit output forms the round key (K_i) for that round.

56 bits $\rightarrow$ 48 bits

- One unique round key is generated for each of the 16 rounds:

$K_1, K_2, K_3, \dots, K_{16}$

- These round keys are used by the DES round function during encryption and decryption.

Advanced Encryption Standard (AES)

AES is a **symmetric key block cipher** used worldwide for secure data encryption. It replaced DES due to its stronger security.

Feature	Description
Type	Symmetric key block cipher
Block Size	128 bits
Key Sizes	128, 192, 256 bits
Structure	Substitution–Permutation Network (NOT Feistel)
Rounds	10, 12, 14 (depending on key size)

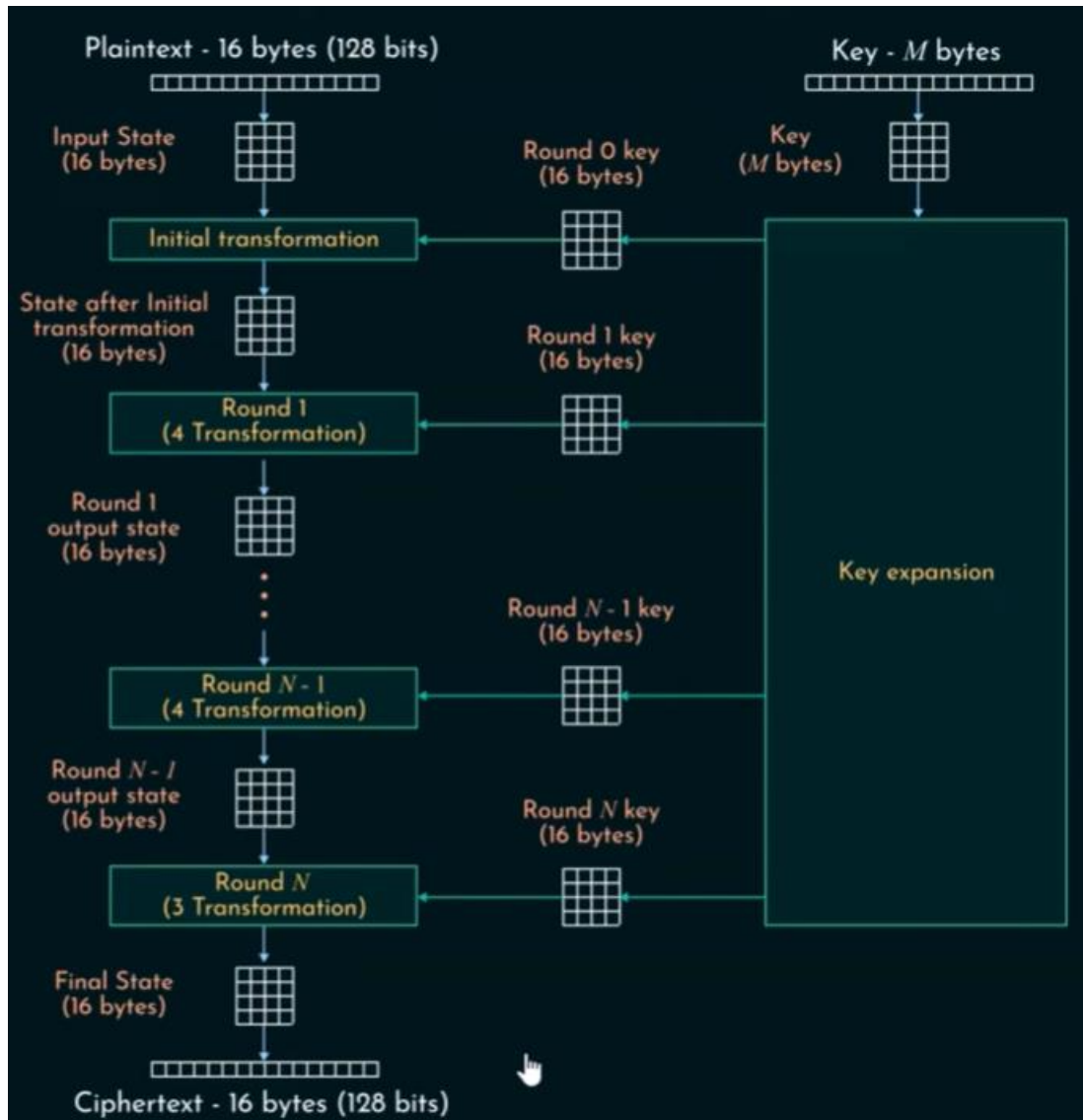
AES Block Structure

- Input block = **128 bits**
- Represented as a **4 × 4 matrix (State matrix)**

B0	B1	B2	B3
B4	B5	B6	B7
B8	B9	B10	B11
B12	B13	B14	B15

Key Size (M)	No of Rounds (N)	Round Key size	Plain Text Size
128-bit	10 rounds	128-bit (16 byte)	128-bit (16 byte)
192-bit	12 rounds	128-bit (16 byte)	128-bit (16 byte)
256-bit	14 rounds	128-bit (16 byte)	128-bit (16 byte)

AES Encryption Process



1) Plaintext Input

- AES takes a **128-bit plaintext block (16 bytes)** as input.
- The plaintext is arranged into a **4 × 4 State Matrix**.

2) Key Expansion

- The secret key (128, 192, or 256 bits) is expanded into multiple **Round Keys**.
- One round key is used in each round of encryption.

3) Initial Transformation (Round 0)

- The plaintext state is XORed with the initial round key.
 $State = State \oplus RoundKey_0$
- This step is called **AddRoundKey**.

4) Round 1 to Round (N-1)

Each round consists of the following four transformations:

a) SubBytes

- Each byte in the State Matrix is replaced using the AES S-Box.
- Provides **confusion**.

b) ShiftRows

- Rows of the State Matrix are shifted left.
- Provides **diffusion**.

Row	Shift
Row 0	0
Row 1	1
Row 2	2
Row 3	3

c) MixColumns

- Each column is transformed using matrix multiplication in $GF(2^8)$.
- Further increases diffusion.

d) AddRoundKey

- The current state is XORed with the round key.
 $State = State \oplus RoundKey_i$

6) Ciphertext Output

- The final State Matrix is converted back into a **128-bit ciphertext block**.
 $Ciphertext = Final\ State$

5) Final Round (Round N)

The final round consists of only **three transformations**:

a) SubBytes

b) ShiftRows

c) AddRoundKey

- **MixColumns** is omitted in the final round.

AES Encryption Flow (Short Form)

1. Input 128-bit plaintext.
2. Arrange plaintext into State Matrix.
3. Perform Key Expansion.
4. Apply initial AddRoundKey.
5. Perform SubBytes.
6. Perform ShiftRows.
7. Perform MixColumns.
8. Perform AddRoundKey.
9. Repeat Steps 5–8 for all intermediate rounds.
10. In the final round perform:
 - SubBytes
 - ShiftRows
 - AddRoundKey
11. Generate 128-bit ciphertext.

Advantages of AES

- DES has fixed no of 16 encryption rounds but AES has N rounds.
- Very strong security
- Large key sizes (128/192/256 bits)
- Resistant to brute-force & Statistical attacks

- Efficient in hardware and software
- Cheap hardware
- Widely adopted globally

Disadvantages

- More complex than DES
- Requires more computational resources

Feature	DES (Data Encryption Standard)	AES (Advanced Encryption Standard)
Cipher Type	Symmetric Block Cipher	Symmetric Block Cipher
Structure	Feistel Network	Substitution-Permutation Network (SPN)
Block Size	64 bits	128 bits
Key Size	56 bits (effective)	128, 192, or 256 bits
Number of Rounds	16	10, 12, or 14
Round Key Size	48 bits	128 bits
Security	Less secure; vulnerable to brute-force attacks	Highly secure and widely used
Speed	Slower	Faster
Confusion & Diffusion	Achieved using Feistel rounds, S-boxes, and permutations	Achieved using SubBytes, ShiftRows, and MixColumns
Encryption & Decryption	Same structure with reverse round keys	Different inverse operations used for decryption
Key Scheduling	Generates 16 round keys	Generates round keys through Key Expansion
Resistance to Attacks	Vulnerable due to small key size	Strong resistance to known attacks
Computational Efficiency	Lower	Higher
Current Usage	Mostly obsolete	Widely used in modern applications
Applications	Legacy systems	VPNs, Wi-Fi, SSL/TLS, Banking, Cloud Security

International Data Encryption Algorithm (IDEA)

The **International Data Encryption Algorithm (IDEA)** is a symmetric-key block cipher designed as a replacement for DES and provides stronger security through the use of multiple mathematical operations.

Feature	Description
Cipher Type	Symmetric Block Cipher
Block Size	64 bits
Key Size	128 bits
Number of Rounds	8 rounds + 1 Output Transformation
Subkeys Generated	52 subkeys
Subkey Size	16 bits each
Structure	Substitution-Permutation Network (SPN)
Year Introduced	1991

Basic Principle

IDEA achieves security by combining three different operations:

1. Bitwise XOR ($\oplus$)
2. Addition modulo 2^{16}
3. Multiplication modulo $(2^{16}+1)$

These operations belong to different algebraic groups, making cryptanalysis extremely difficult.

Block Division

- Plaintext block = 64 bits
- Divided into four 16-bit sub-blocks:
 X_1, X_2, X_3, X_4
- Each sub-block: 16 bits

Advantages of IDEA

1. Strong security.
2. Resistant to differential and linear cryptanalysis.
3. Uses a large 128-bit key.
4. No practical attacks on the full IDEA algorithm.
5. Combines three different mathematical operations for enhanced confusion and diffusion.

Key Scheduling

- User provides a **128-bit key**.
- The key schedule generates: 52 subkey
- Each subkey is: 16 bits

Distribution of Subkeys

- Rounds 1–8 require 6 subkeys each $8 \times 6 = 48$
- Output transformation requires 4 subkeys $48 + 4 = 52$
- Thus, total subkeys: 52

Disadvantages of IDEA

1. More computationally intensive than some modern ciphers.
2. Complex key scheduling.
3. Slower than AES in software implementations.
4. Largely replaced by AES in modern applications.

IDEA Encryption Process

Step 1: Input Plaintext

- Take a 64-bit plaintext block.
- Divide into four 16-bit blocks: X_1, X_2, X_3, X_4

Step 2: Perform 8 Rounds

Each round uses six subkeys: $K_1, K_2, K_3, K_4, K_5, K_6$

Round Operations

- Multiplication:** $X_1 = X_1 \times K_1 \pmod{2^{16} + 1}$
- Addition:** $X_2 = X_2 + K_2 \pmod{2^{16}}$
- Addition:** $X_3 = X_3 + K_3 \pmod{2^{16}}$
- Multiplication:** $X_4 = X_4 \times K_4 \pmod{2^{16} + 1}$
- XOR Operations:** Intermediate values are generated using XOR operations.
- Additional Multiplication and Addition** Subkeys K_5 and K_6 are used.
- Swap:** $X_2 \leftrightarrow X_3$. The second and third blocks are swapped after every round except the final round.

Step 4: Generate Ciphertext

- The resulting four 16-bit blocks are combined to form: 64-bit Ciphertext

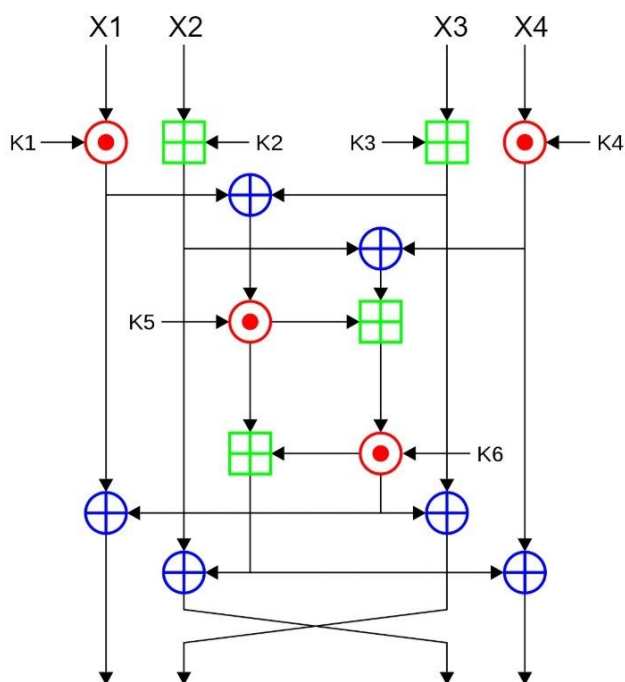
Step 3: Output Transformation

After Round 8, a final transformation is performed using four remaining subkeys.

Operations:

- Multiplication
- Addition
- Addition
- Multiplication

using subkeys: $K_{49}, K_{50}, K_{51}, K_{52}$



Where,



= Modular Addition



= Modular Multiplication



= Bitwise XOR

1. Input 64-bit plaintext.
2. Divide plaintext into four 16-bit blocks.
3. Generate 52 subkeys from the 128-bit key.
4. Perform 8 rounds of encryption using multiplication, addition, and XOR operations.
5. Swap middle two blocks after each round.
6. Perform final output transformation using four remaining subkeys.
7. Combine the four 16-bit blocks.
8. Obtain the 64-bit ciphertext.

IDEA Round Structure

For each round:

1. Multiply X_1 with K_1
2. Add X_2 with K_2
3. Add X_3 with K_3
4. Multiply X_4 with K_4
5. XOR intermediate values
6. Multiply using K_5
7. Add using K_6
8. XOR operations
9. Swap X_2 and X_3

Decryption in IDEA

- Uses the same algorithm structure.
- Requires specially generated inverse subkeys.
- Operations are reversed using:
 - Multiplicative inverse modulo $(2^{16}+1)$
 - Additive inverse modulo 2^{16}

Important Formulae

- Addition: $A + B \pmod{2^{16}}$
- Multiplication: $A \times B \pmod{(2^{16} + 1)}$
- XOR: $A \oplus B$

RC5 (Rivest Cipher 5)

RC5 is a **symmetric-key block cipher** designed by **Ron Rivest** in 1994. It is known for its simplicity, flexibility, and parameterized design.

Features of RC5

Feature	Description
Cipher Type	Symmetric Block Cipher
Structure	Feistel-like Network
Block Size	Variable (32, 64, 128 bits)
Key Size	Variable (0–2040 bits)
Number of Rounds	Variable (0–255 rounds)
Operations Used	XOR, Addition, Circular Rotation

RC5 Parameters

RC5 is represented as: RC5 – w/r/b . RC5-32/12/16, you say: "RC5 with 32-bit words, 12 rounds, and a 16-byte key.

- w= Word size (16, 32, or 64 bits)
- r= Number of rounds
- b= Key length in bytes

RC5 Block Structure

A plaintext block consists of two words: A, B

Each word has w bits.

For RC5-32:

$$A = 32 \text{ bits}$$

$$B = 32 \text{ bits}$$

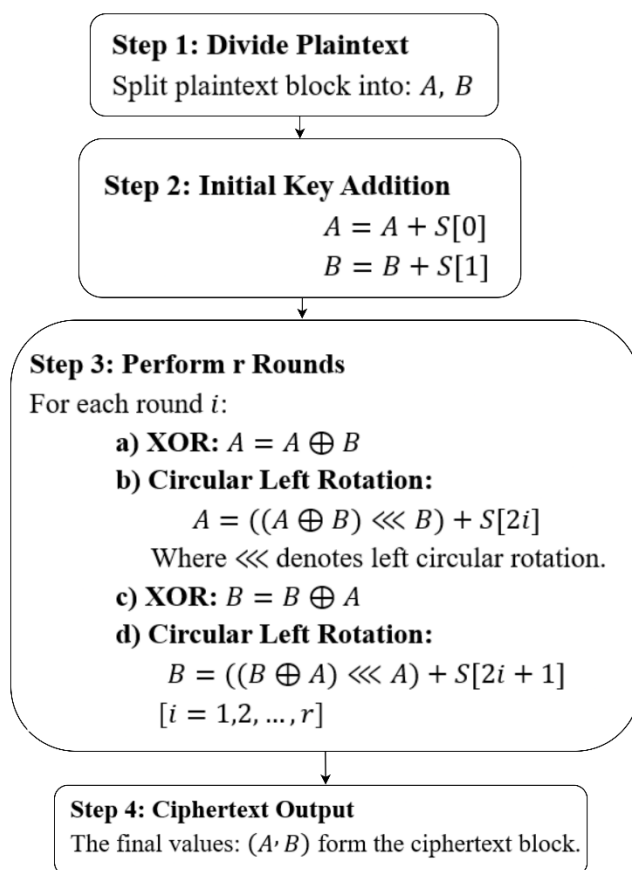
Therefore, *Block Size* = 64 bits

Key Expansion

- User provides a secret key.
- The key expansion algorithm generates: $S[0], S[1], S[2], \dots, S[2r + 1]$

For r rounds: $2r + 2$ subkeys are generated. **Example:** For 12 rounds: $2(12) + 2 = 26$ subkeys.

RC5 Encryption Process



RC5 Encryption Flow

1. Input plaintext.
2. Split into two words A and B .
3. Add first two subkeys.
4. XOR A with B .
5. Rotate left.
6. Add round key.
7. XOR B with A .
8. Rotate left.
9. Add round key.
10. Repeat for all rounds.
11. Obtain ciphertext.

RC5 Decryption

- Decryption performs the inverse operations.
- Round keys are used in reverse order.
- Rotations are reversed using right circular shifts.

Operations Used in RC5

- 1) **XOR:** $A \oplus B$ Provides confusion.
- 2) **Modular Addition:** $A + B \pmod{2^w}$ Introduces non-linearity.
- 3) **Circular Rotation** Left Rotation: $A \lll n$. Example: $10110001 \rightarrow 11000110$. Provides diffusion.

Advantages of RC5

1. Simple design.
2. Fast implementation.
3. Variable block size.
4. Variable key size.

5. Variable number of rounds.
6. Suitable for hardware and software implementations.

1. Patented algorithm (historically).
2. Less commonly used today compared to AES.
3. Security depends on chosen parameters.

Disadvantages of RC5

RC5 vs DES

Feature	RC5	DES
Block Size	Variable	64 bits
Key Size	Variable	56 bits
Rounds	Variable	16
Operations	XOR, Addition, Rotation	Feistel Operations
Flexibility	High	Low
Security	Stronger	Weaker